

Cumulative Constraint

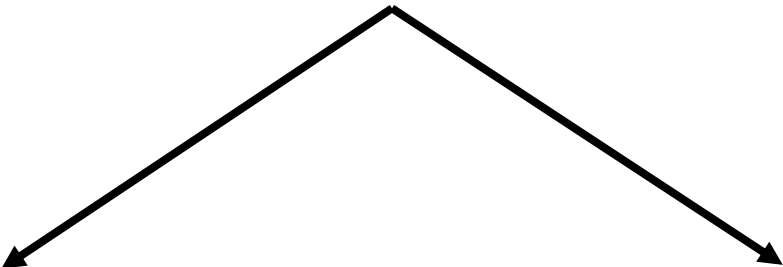
Modern Constraint Solving
(ESSAI'26)

Emir Demirović



In this mini-lecture...

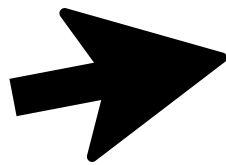
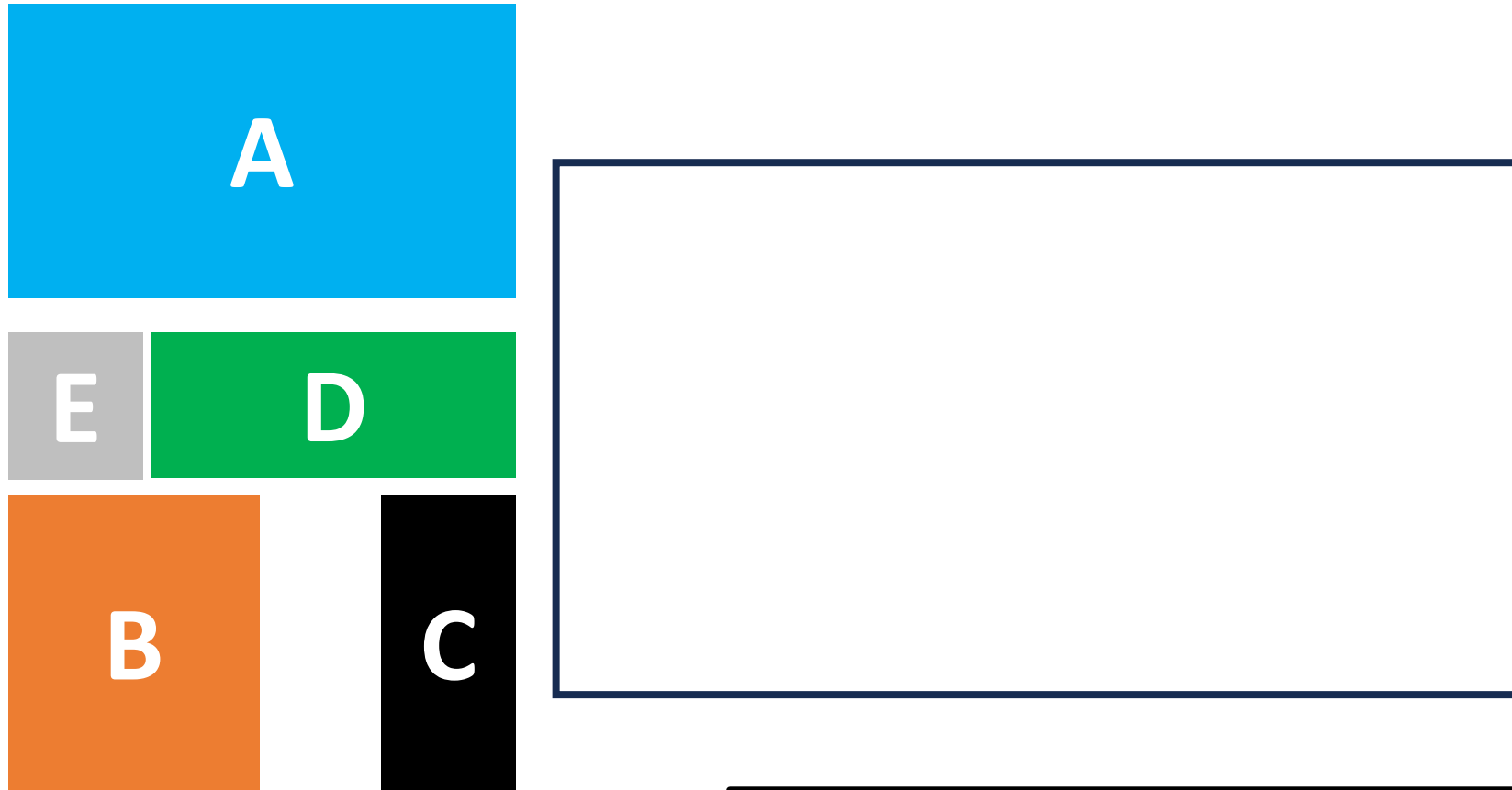
Cumulative Constraint



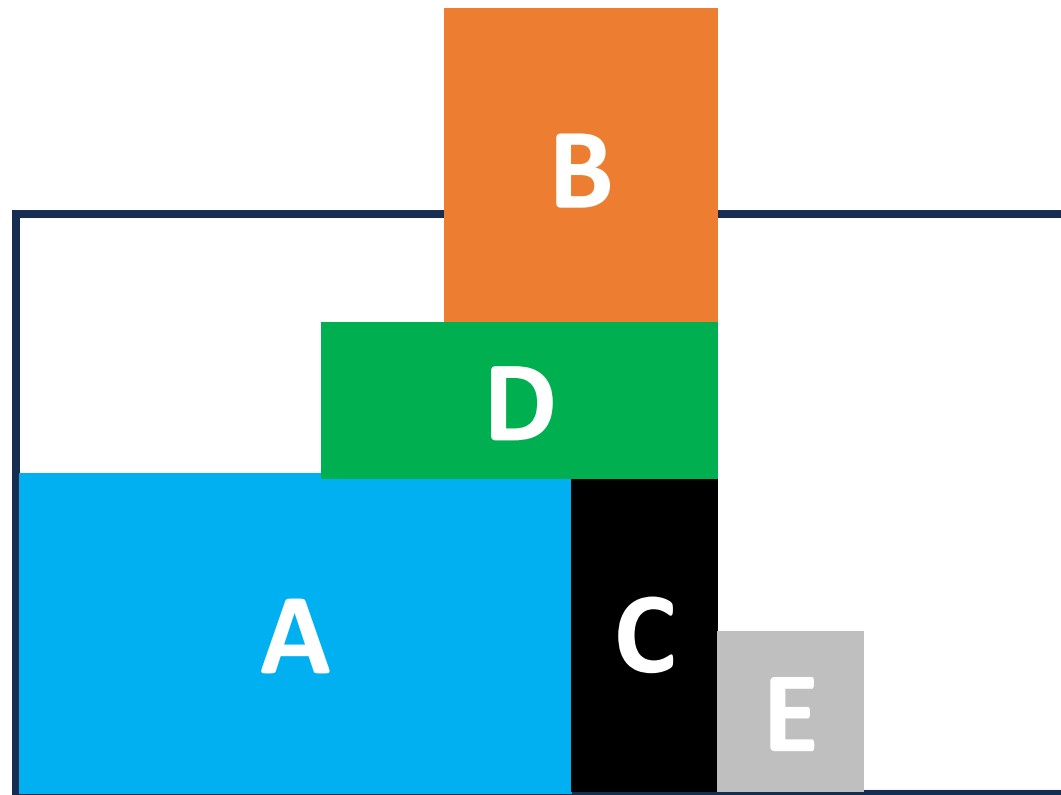
```
graph TD; A[Cumulative Constraint] --> B[Timetable Reasoning]; A --> C[Energetic Reasoning];
```

**Timetable
Reasoning**

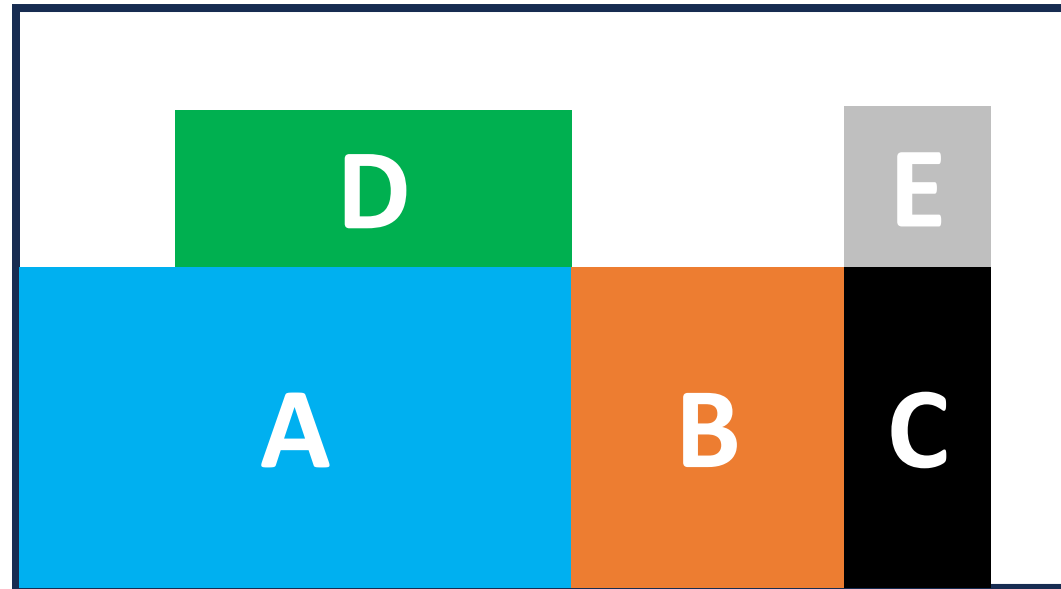
**Energetic
Reasoning**



**Can the smaller rectangles
be placed within
the large white rectangle?**



Infeasible!



Feasible!

Cumulative Constraint

$$task_i = (s_i, D_i, R_i)$$



Integer variable representing possible starting times

Cumulative Constraint

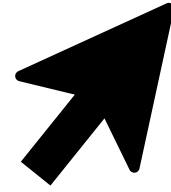
$$task_i = (s_i, D_i, R_i)$$



Constant representing task duration

Cumulative Constraint

$$task_i = (s_i, D_i, R_i)$$

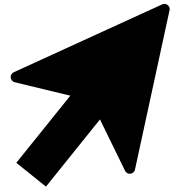


Constant representing resource requirements

Cumulative Constraint

$$task_i = (s_i, D_i, R_i)$$

R_{max}



Constant representing maximum number of available resources

Cumulative Constraint

$$task_i = (s_i, D_i, R_i)$$

$$R_{max}$$

“The resource consumption cannot exceed the maximum at any time”

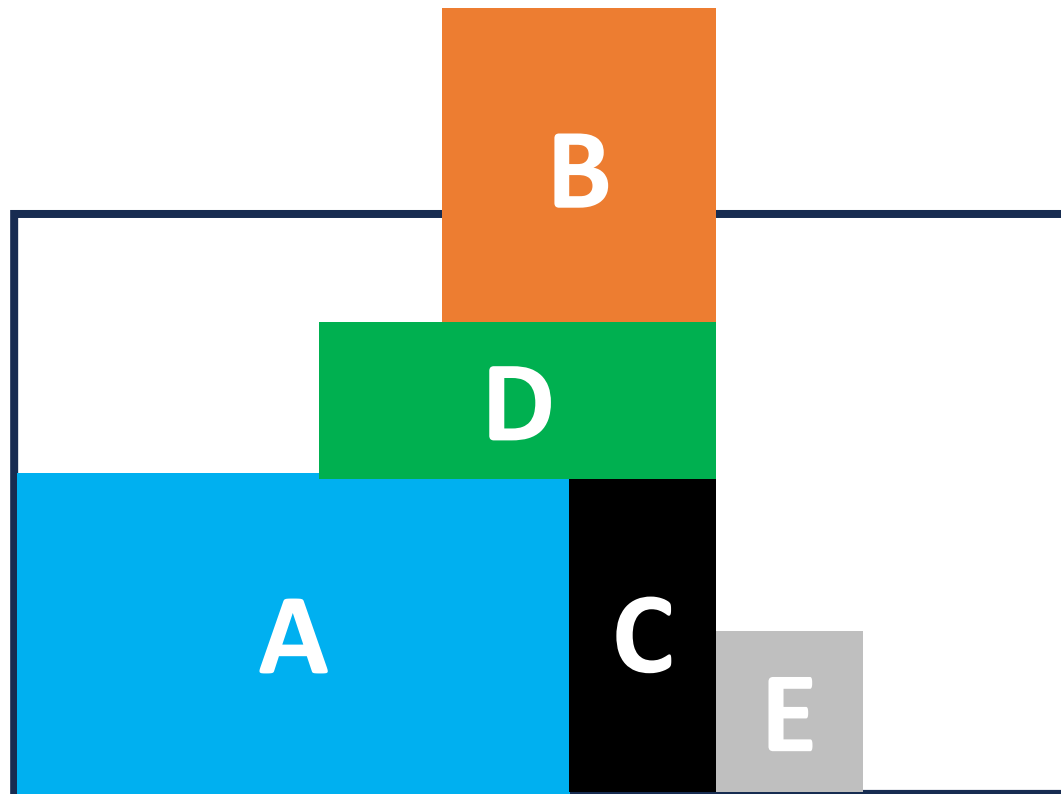
Cumulative Constraint

$$task_i = (s_i, D_i, R_i)$$

$$R_{max}$$

“The resource consumption cannot exceed the maximum at any time”

Infeasible!



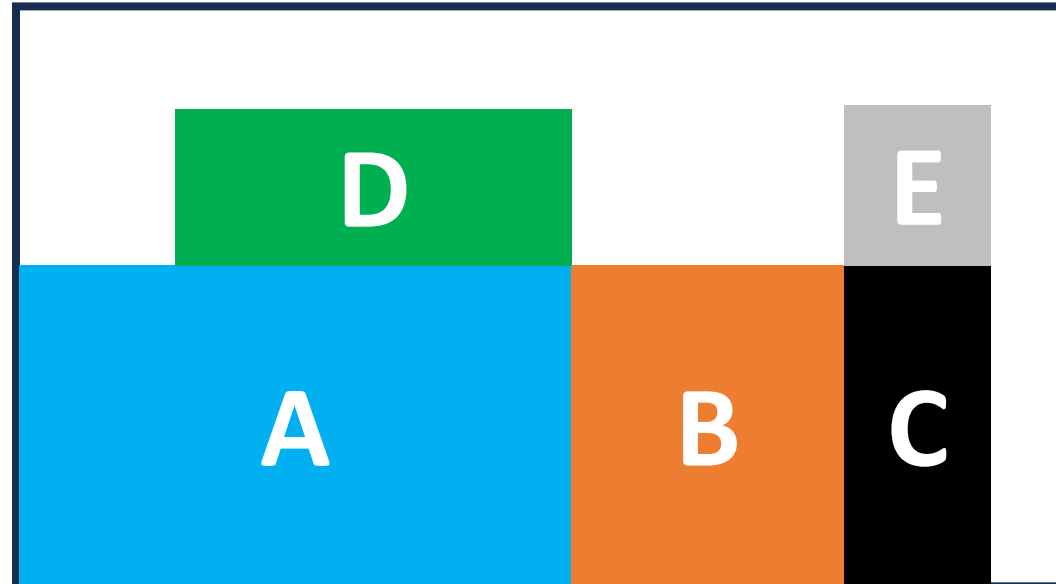
Cumulative Constraint

$$task_i = (s_i, D_i, R_i)$$

$$R_{max}$$

“The resource consumption cannot exceed the maximum at any time”

Feasible!



Cumulative Constraint

$$task_i = (s_i, D_i, R_i)$$

$$R_{max}$$

“The resource consumption cannot exceed the maximum at any time”

Determining the existence of a feasible solution for cumulative is
NP-complete

(resource-constrained scheduling is NP-hard, but so is its subproblem!)

Cumulative Constraint

$$task_i = (s_i, D_i, R_i)$$

$$R_{max}$$

“The resource consumption cannot exceed the maximum at any time”

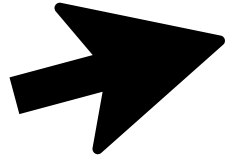
Determining the existence of a feasible solution for cumulative is
NP-complete

...but since it appears as a common subproblem,
let us try to do some propagation in polynomial time

Variant 1:
Timetabling reasoning
for Cumulative

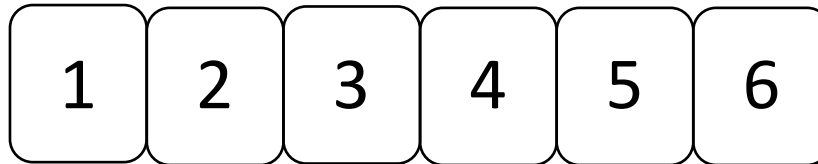
Cumulative Propagator

Integer variable
representing the start time
of the task

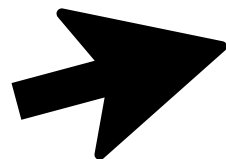


$$s = \{1, 2, 3\}$$

$$D = 4$$

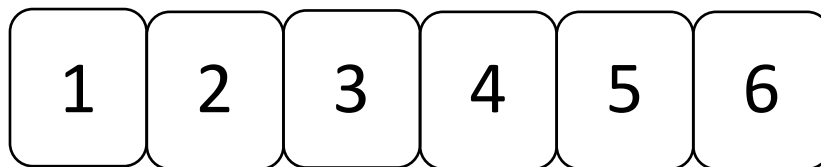


Cumulative Propagator



$$s = \{1, 2, 3\}$$

$$D = 4$$

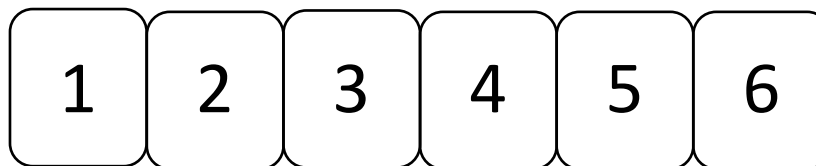
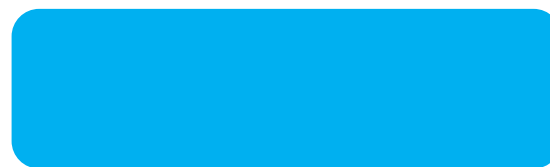
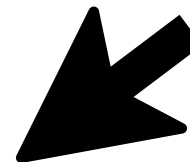


Cumulative Propagator

$$s = \{1, 2, 3\}$$

$$D = 4$$

$$s = 3$$

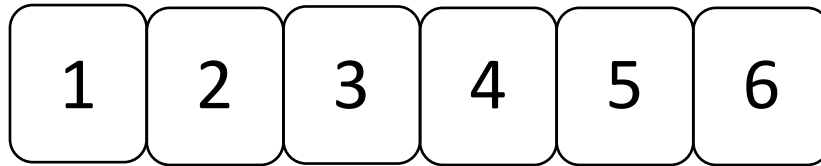
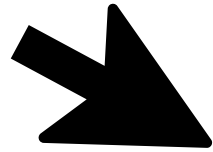


Cumulative Propagator

$$s = \{1, 2, 3\}$$

$$D = 4$$

$$s = 2$$

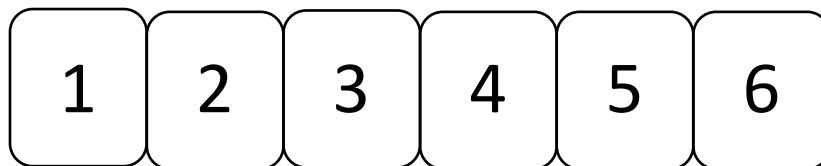
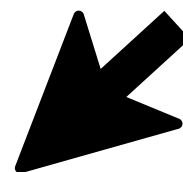


Cumulative Propagator

$$s = \{1, 2, 3\}$$

$$D = 4$$

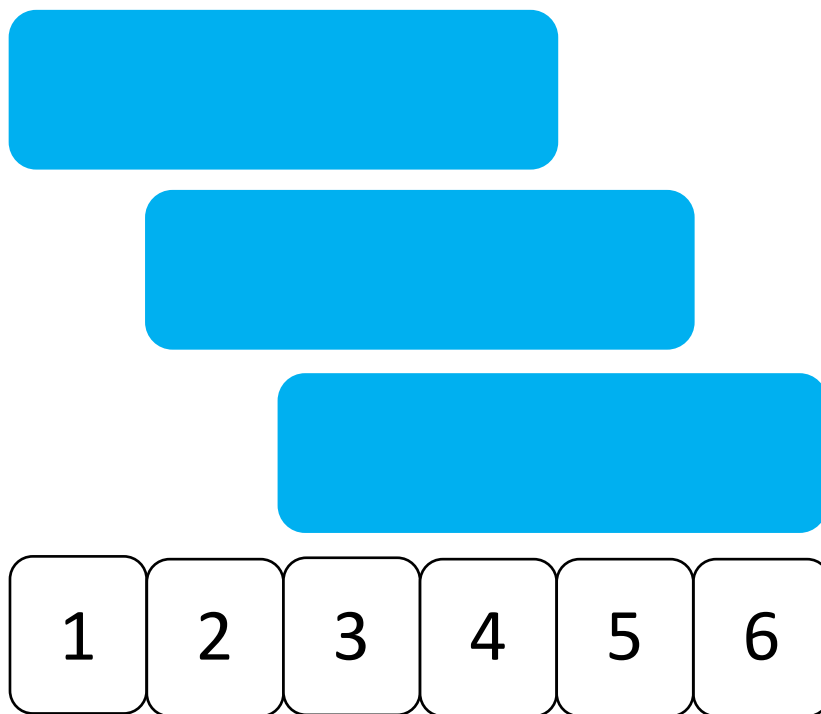
$$s = 1$$



Cumulative Propagator

$$s = \{1, 2, 3\}$$

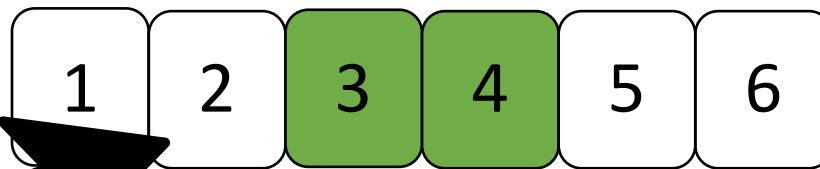
$$D = 4$$



Cumulative Propagator

$$s = \{1, 2, 3\}$$

$$D = 4$$

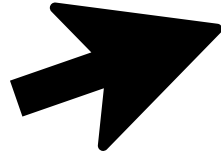


Compulsory
resource consumption

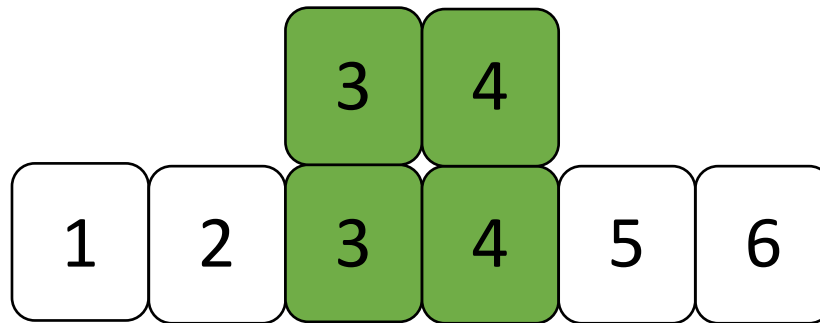
Cumulative Propagator

$$s_1, s_2 = \{1, 2, 3\}$$

$$D_1, D_2 = 4$$



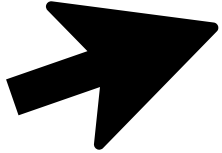
Two tasks

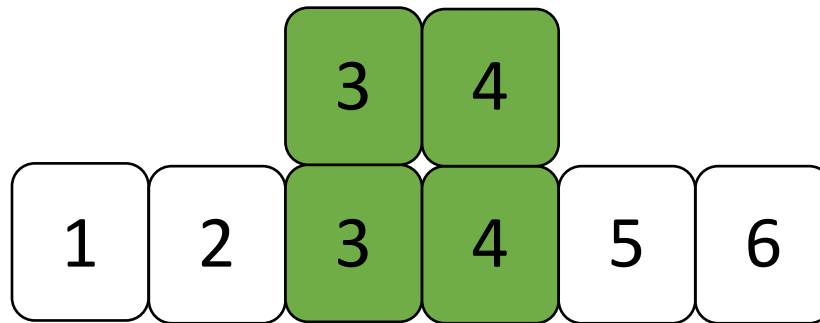


Cumulative Propagator

$$s_1, s_2 = \{1, 2, 3\}$$

$$D_1, D_2 = 4$$


$$R_{max} = 1$$



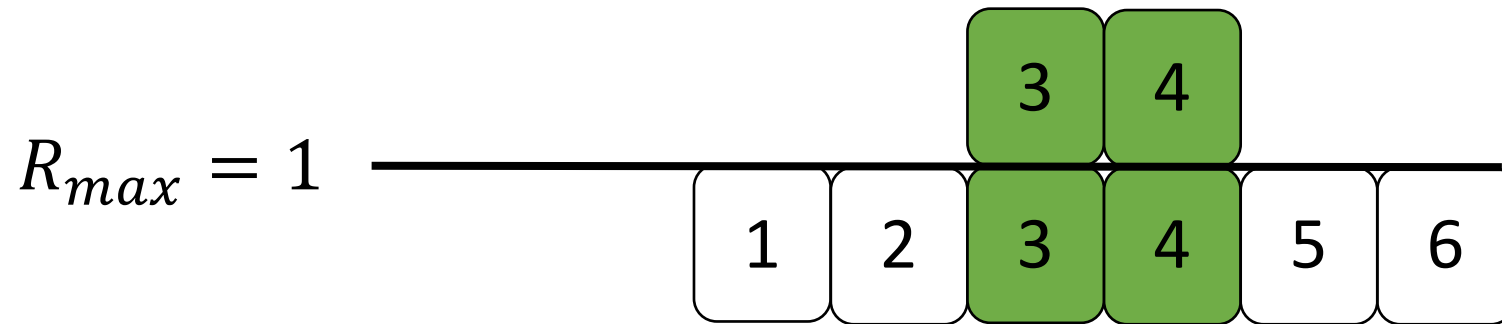
Cumulative Propagator

$$s_1, s_2 = \{1, 2, 3\}$$

$$D_1, D_2 = 4$$

$$R_{max} = 1$$

Compulsory part reasoning reveals infeasibility



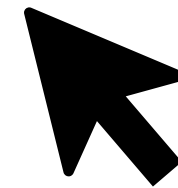
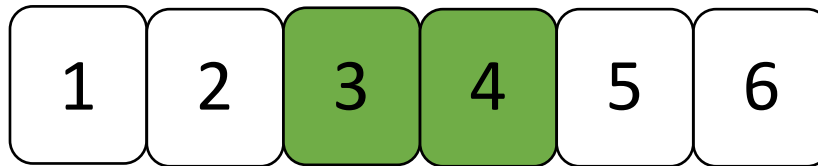
Cumulative Propagator

$$s_1, s_2 = \{1, 2, 3\}$$

$$D_1, D_2 = 4$$

$$R_{max} = 1$$

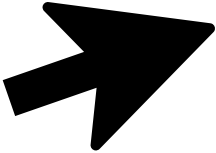
$$f(\{1, 2, 3\}, \{1, 2, 3\}) = (\{\}, \{\})$$



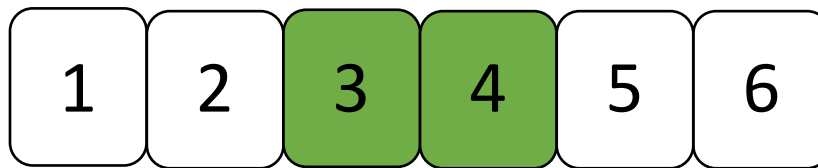
empty domains,
no feasible solution
exists

Cumulative Propagator

$$s_1 = \{1, 2, 3\}, s_2 = \{2, 3, 4, 5\}$$


$$D_1 = 4, D_2 = 2$$

$$R_{max} = 1$$

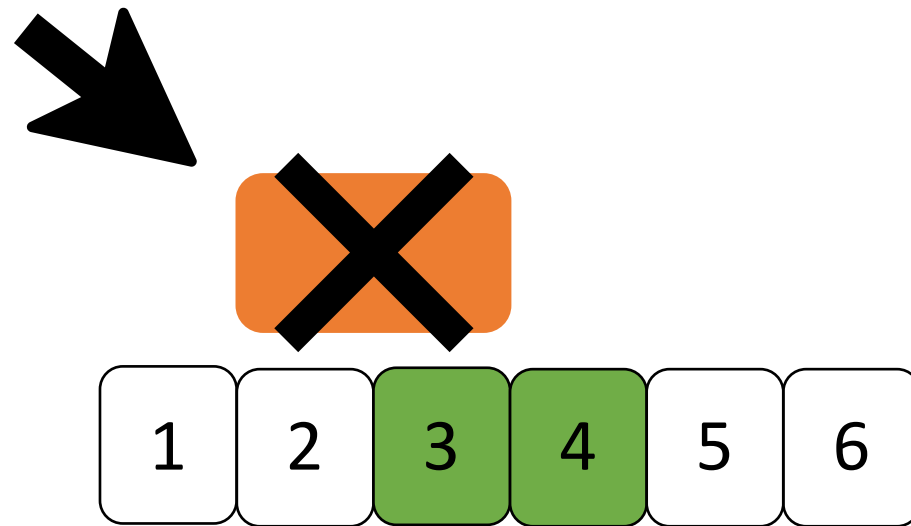


Cumulative Propagator

$$s_1 = \{1, 2, 3\}, s_2 = \{2, 3, 4, 5\}$$

$$D_1 = 4, D_2 = 2$$

$$R_{max} = 1$$



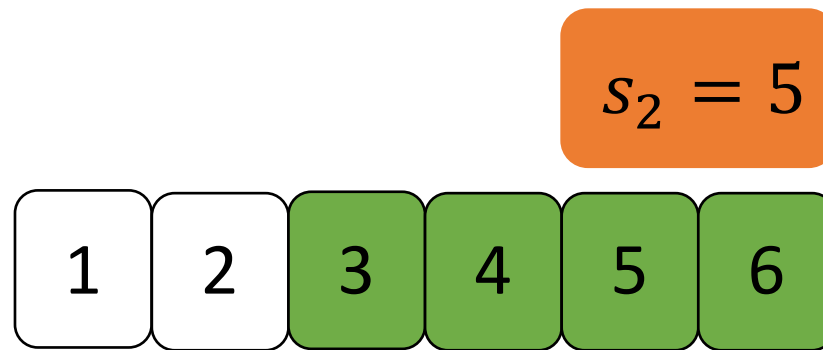
Cumulative Propagator

$$s_1 = \{1, 2, 3\}, s_2 = \{2, 3, 4, 5\}$$

$$D_1 = 4, D_2 = 2$$

$$R_{max} = 1$$

Compulsory part reasoning forces task 2 to start at time 5



Cumulative Propagator

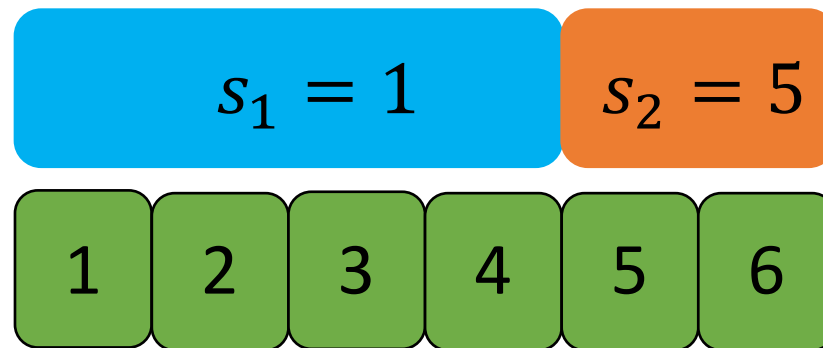
$$s_1 = \{1, \cancel{2}, \cancel{3}\}, s_2 = \{\cancel{2}, \cancel{3}, \cancel{4}, 5\}$$

$$D_1 = 4, D_2 = 2$$

$$R_{max} = 1$$

Compulsory part reasoning reveals the only feasible assignment

“timetable reasoning”



Cumulative Propagator

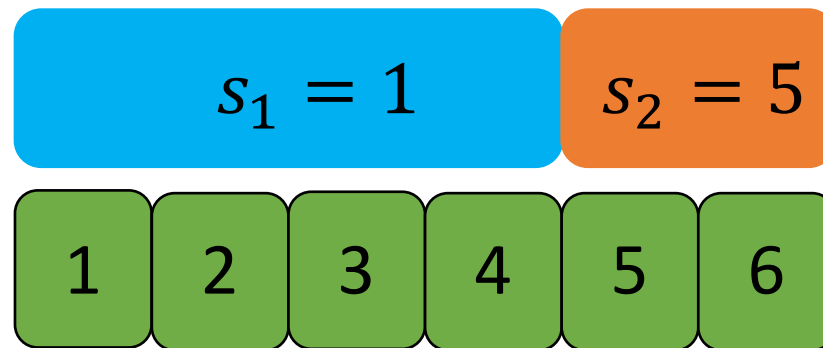
$$s_1 = \{1, \cancel{2}, \cancel{3}\}, s_2 = \{\cancel{2}, \cancel{3}, \cancel{4}, 5\}$$

$$D_1 = 4, D_2 = 2$$

$$R_{max} = 1$$

Compulsory part reasoning reveals the only feasible assignment

“timetable reasoning”

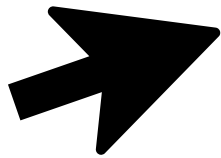


Cumulative Propagator

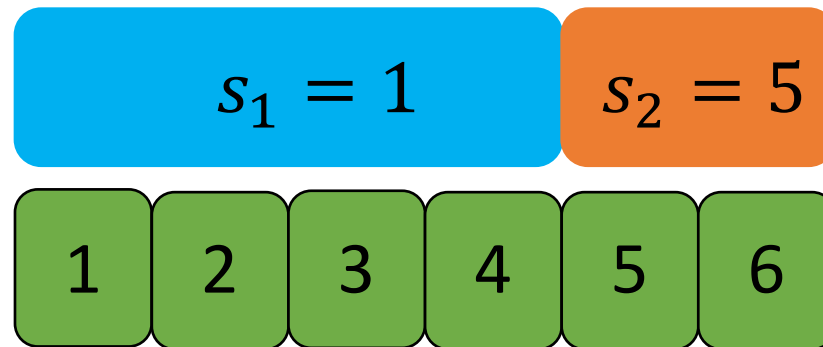
$$s_1 = \{1, 2, 3\}, s_2 = \{2, 3, 4, 5\}$$

$$D_1 = 4, D_2 = 2$$

$$R_{max} = 1$$

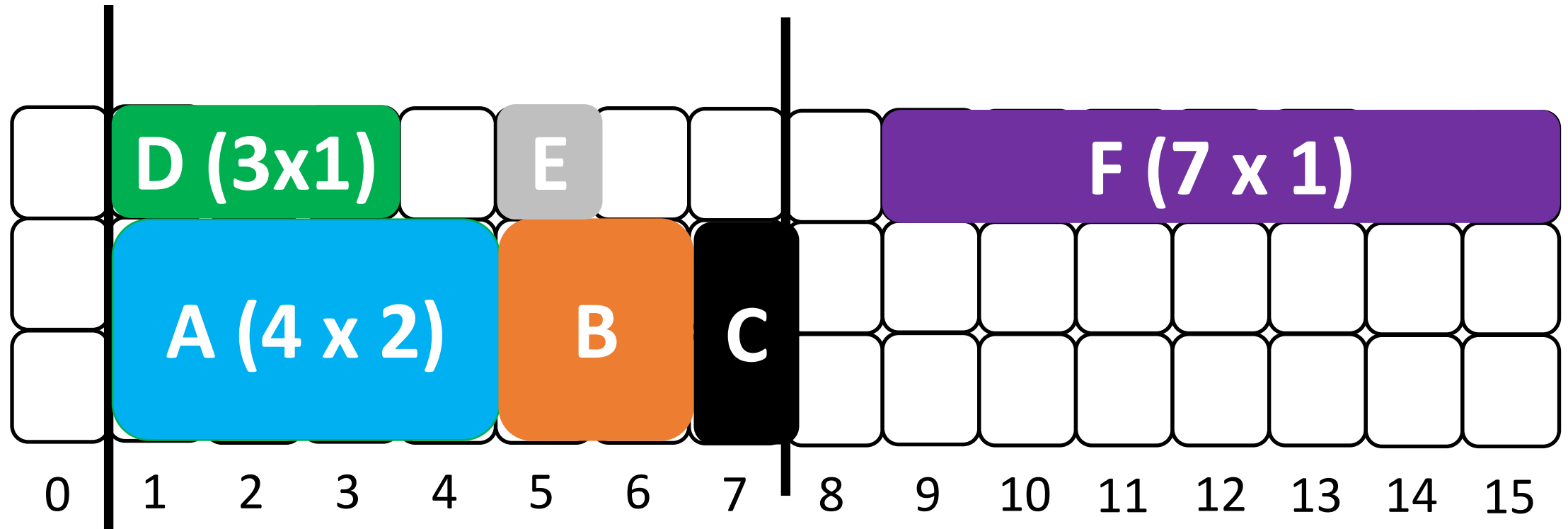


$$f(\{1, 2, 3\}, \{2, 3, 4, 5\}) = (\{1\}, \{5\})$$

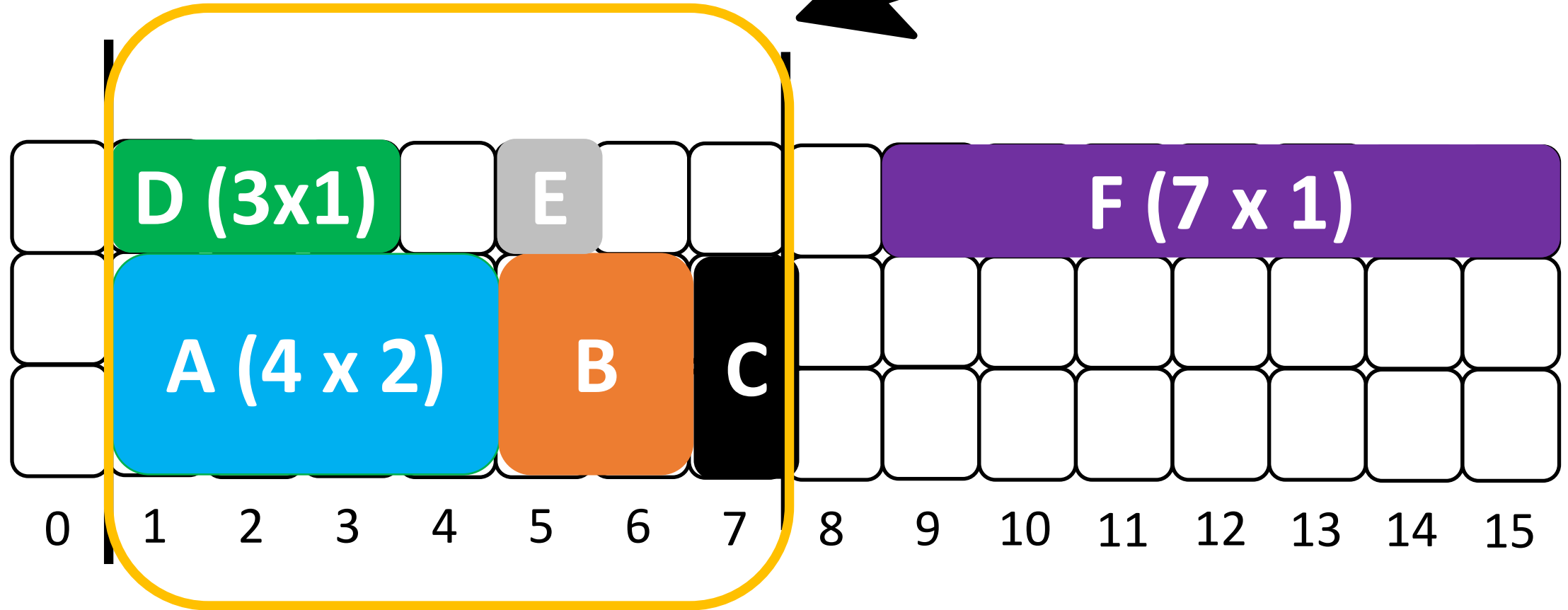
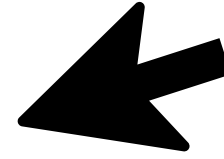


Variant 2:
Energetic Reasoning
for Cumulative

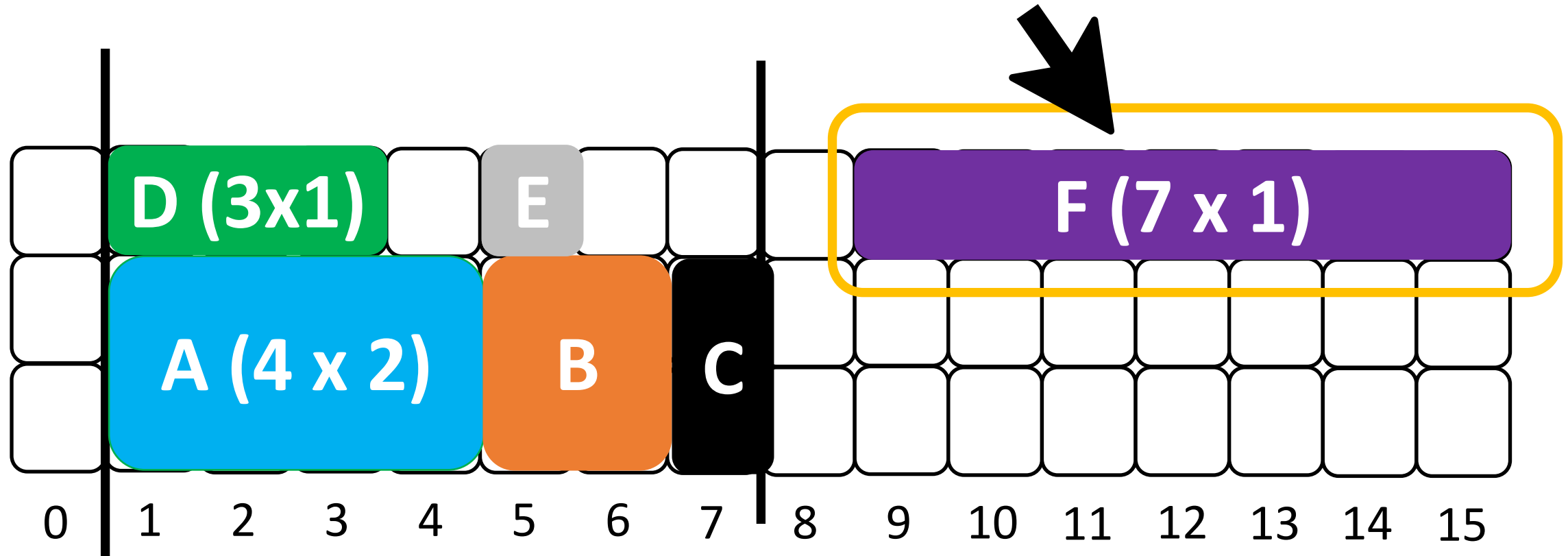
Three units of resource available



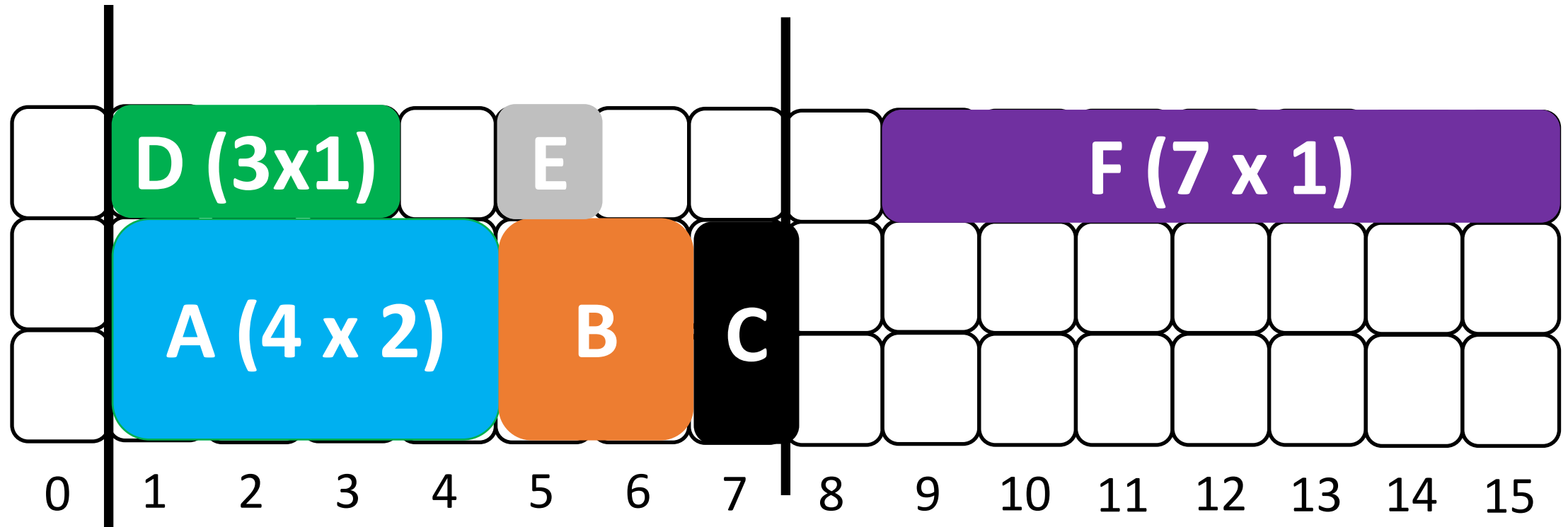
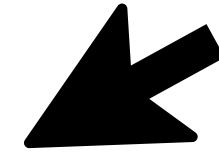
These tasks must complete
within this range 1-7

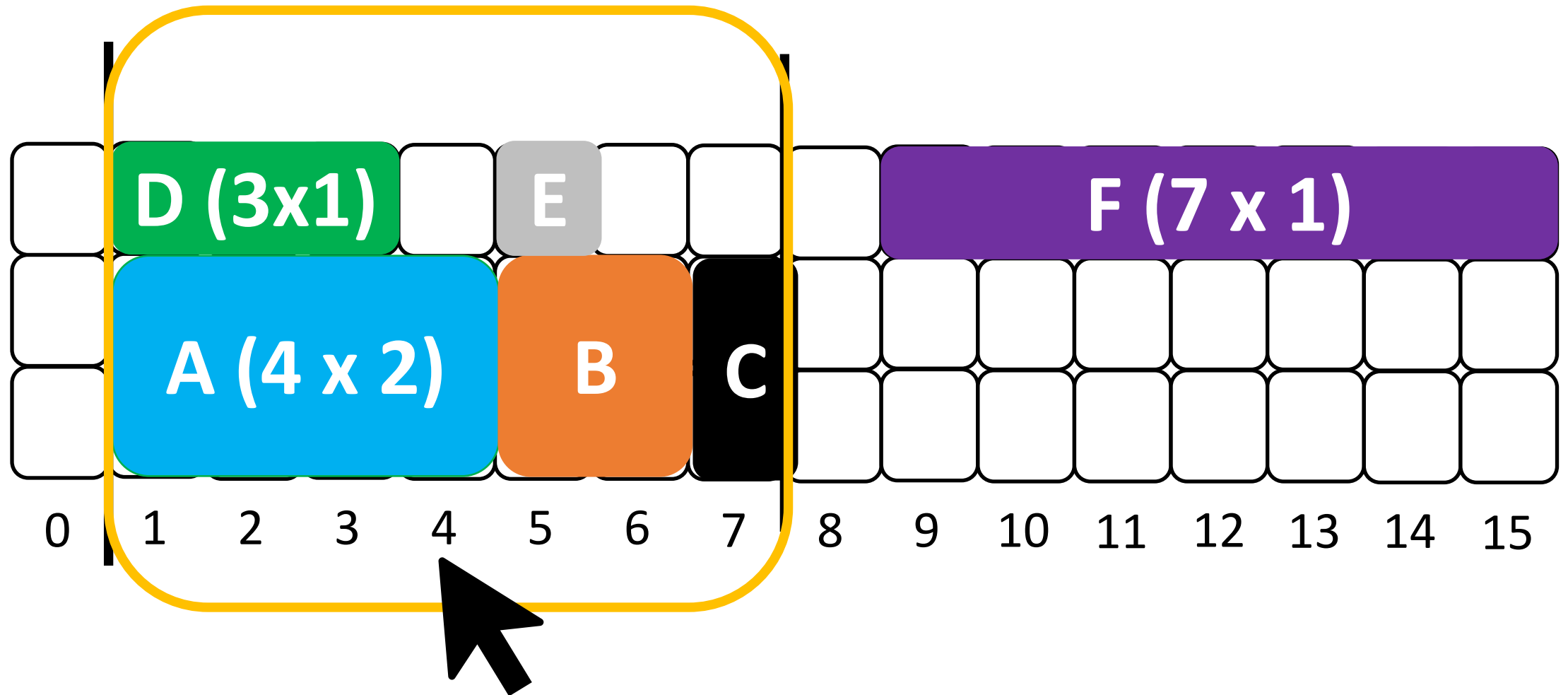


Task F can be completed any time



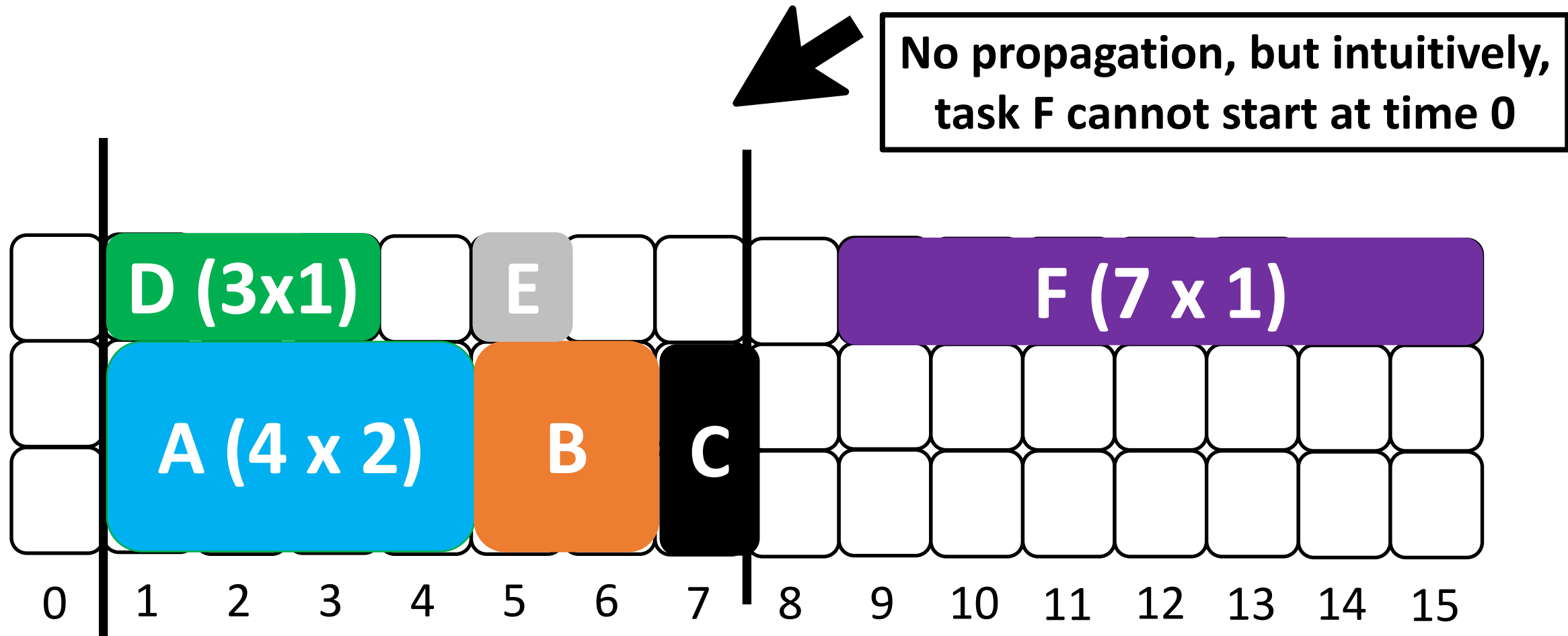
What does timetabling reasoning infer?

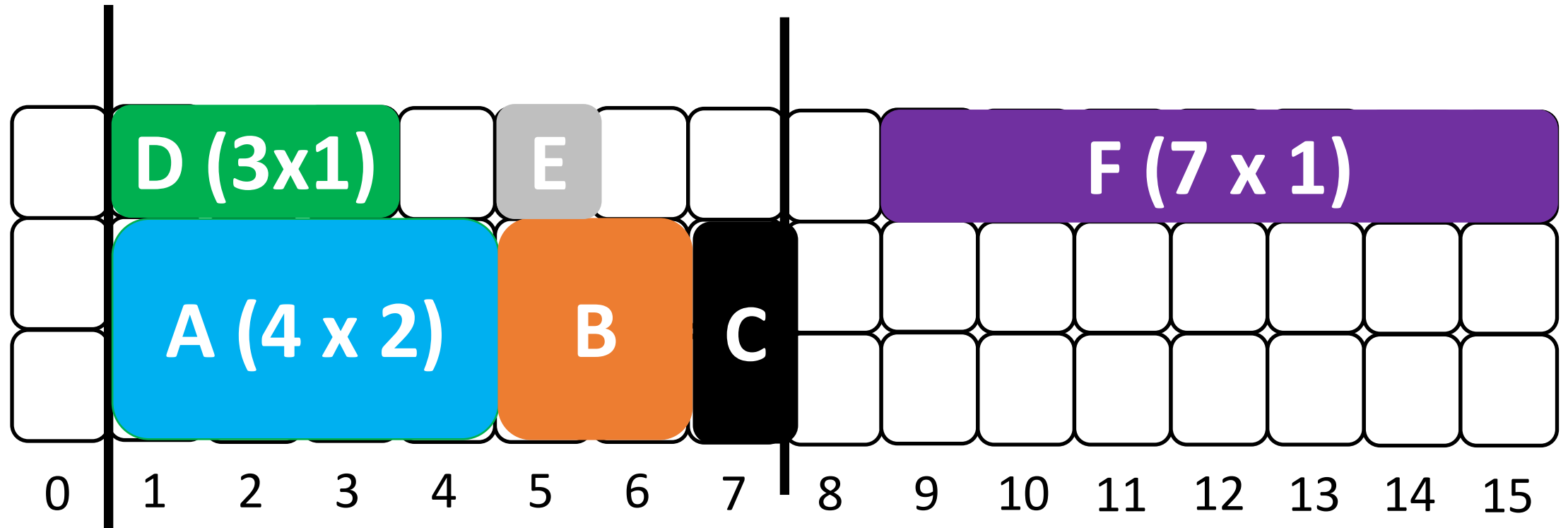




Only Task A has a mandatory part at time 4 → no propagation

Example adapted from A quadratic edge-finding filtering algorithm for cumulative resource constraints– Kameugne et al. (2013)





Need something different than timetabling!

Energetic reasoning

A (4 x 2)

D (3x1)

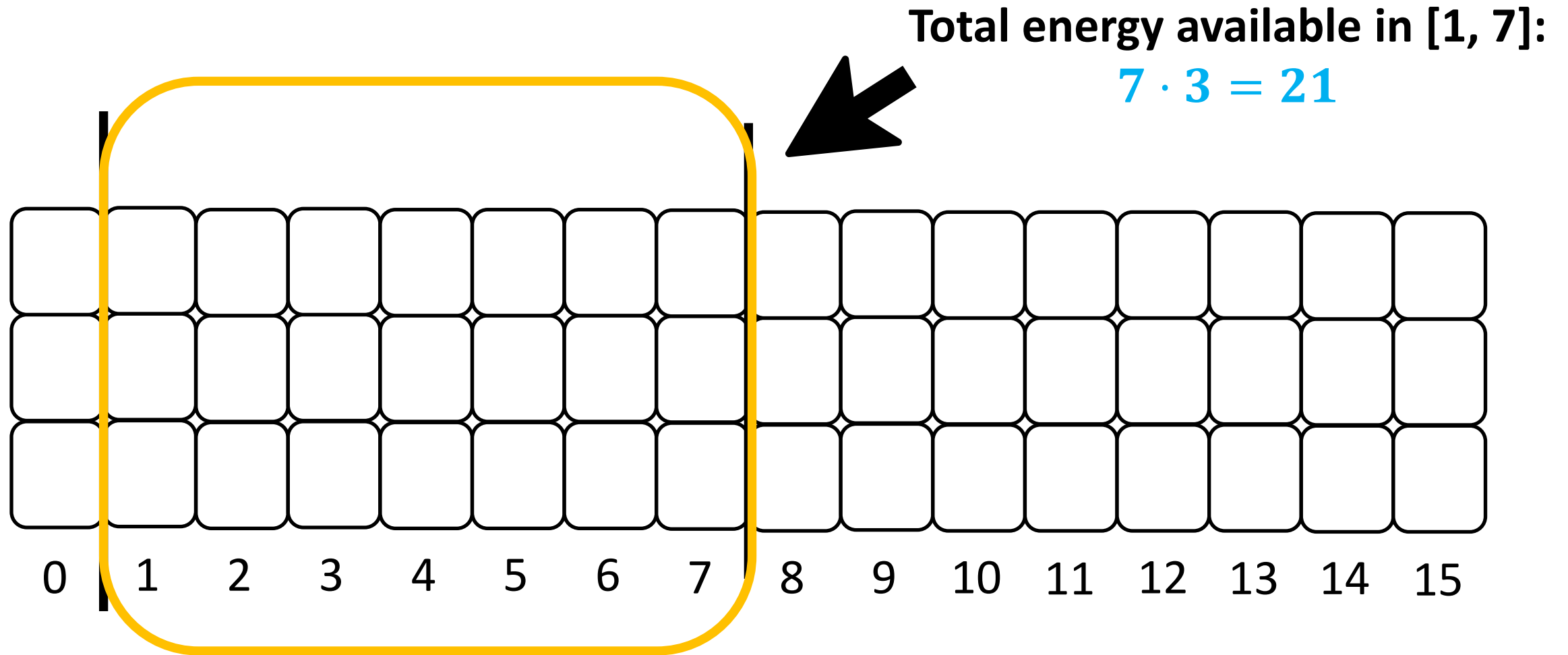
B

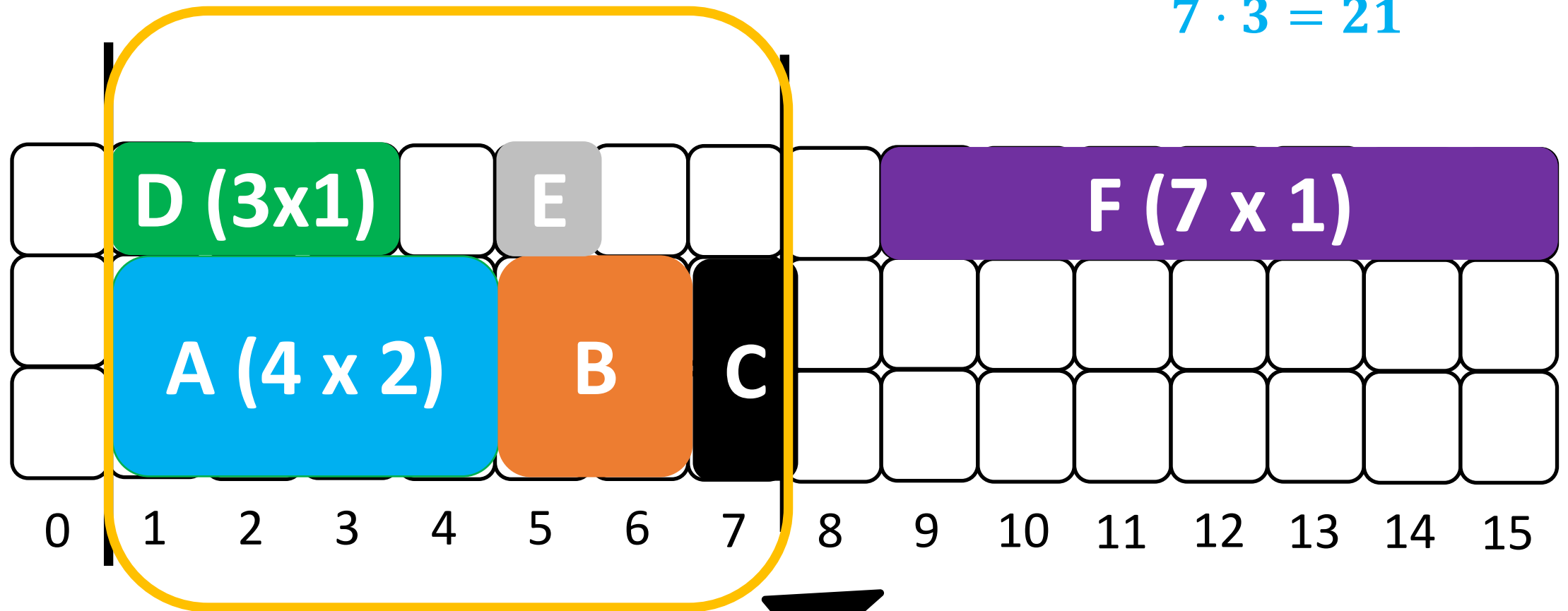
8

3

4

$$\text{energy}(\text{task}) = \text{duration} \cdot \text{resource_demand}$$





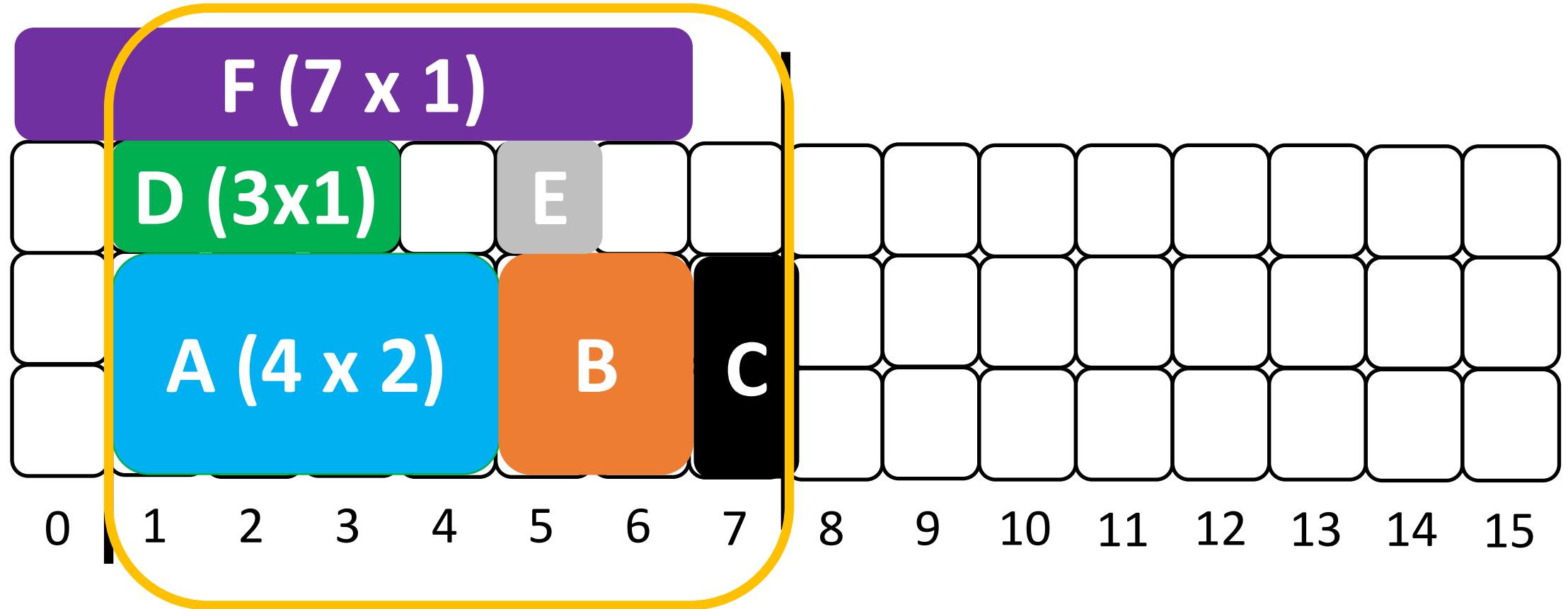
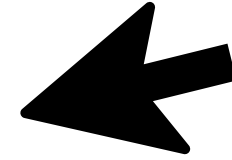
Total energy available in [1, 7]:

$$7 \cdot 3 = 21$$

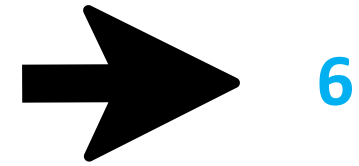
Energy needed for the tasks:

$$8 + 4 + 2 + 3 + 1 = 18$$

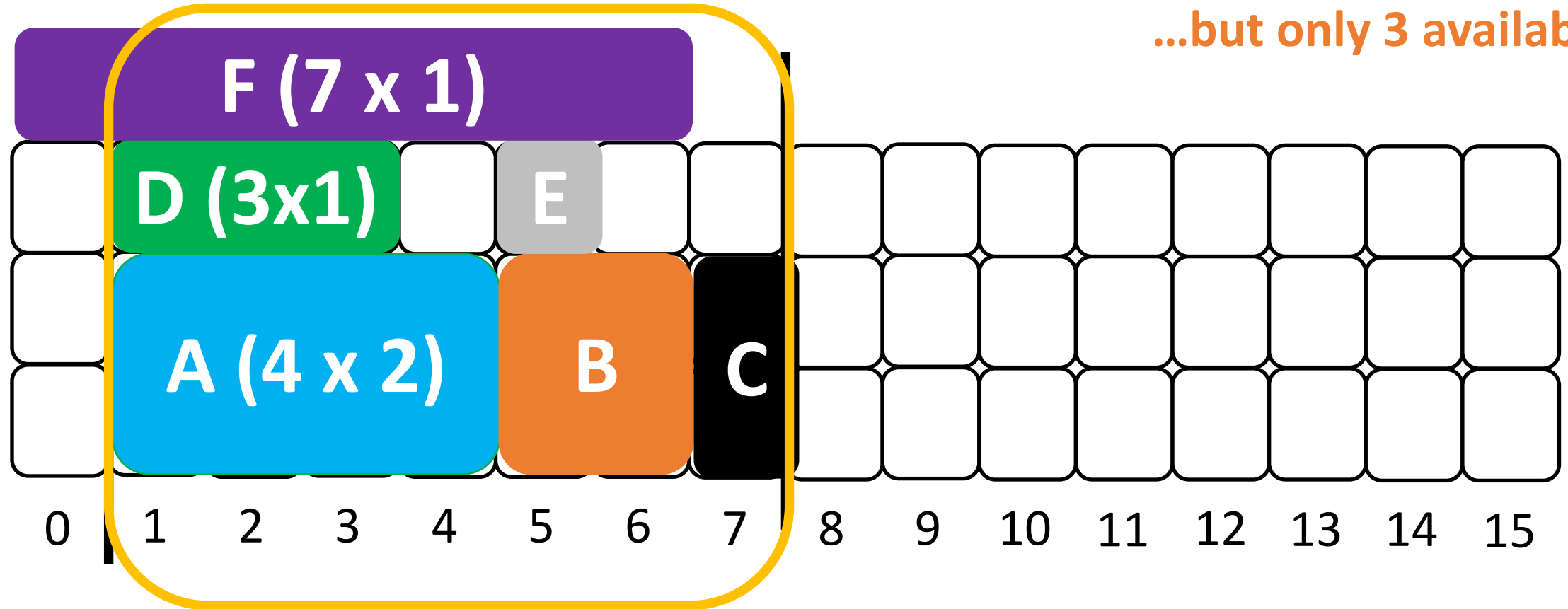
Place task F at its earliest starting time...
How much energy would task F need from [1, 7]?



Place task F at its earliest starting time...
How much energy would task F need from [1, 7]?



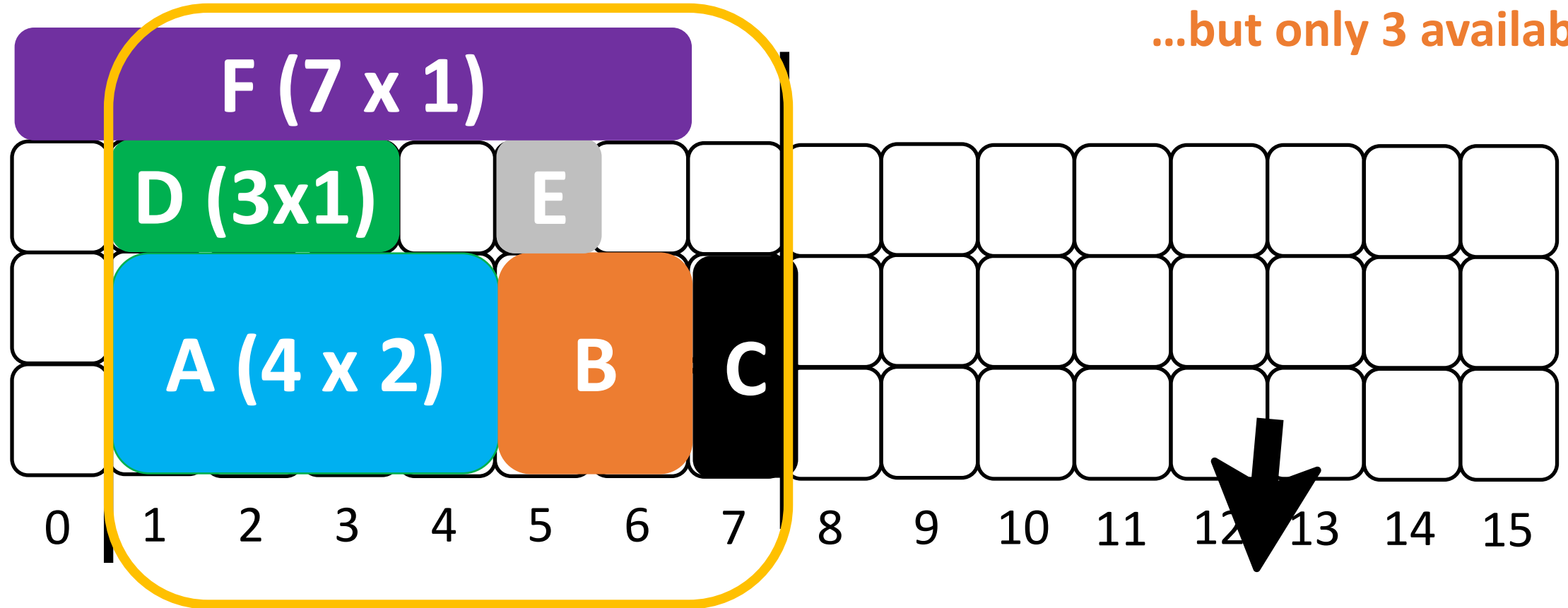
...but only 3 available



Place task F at its earliest starting time...
How much energy would task F need from [1, 7]?

6

...but only 3 available

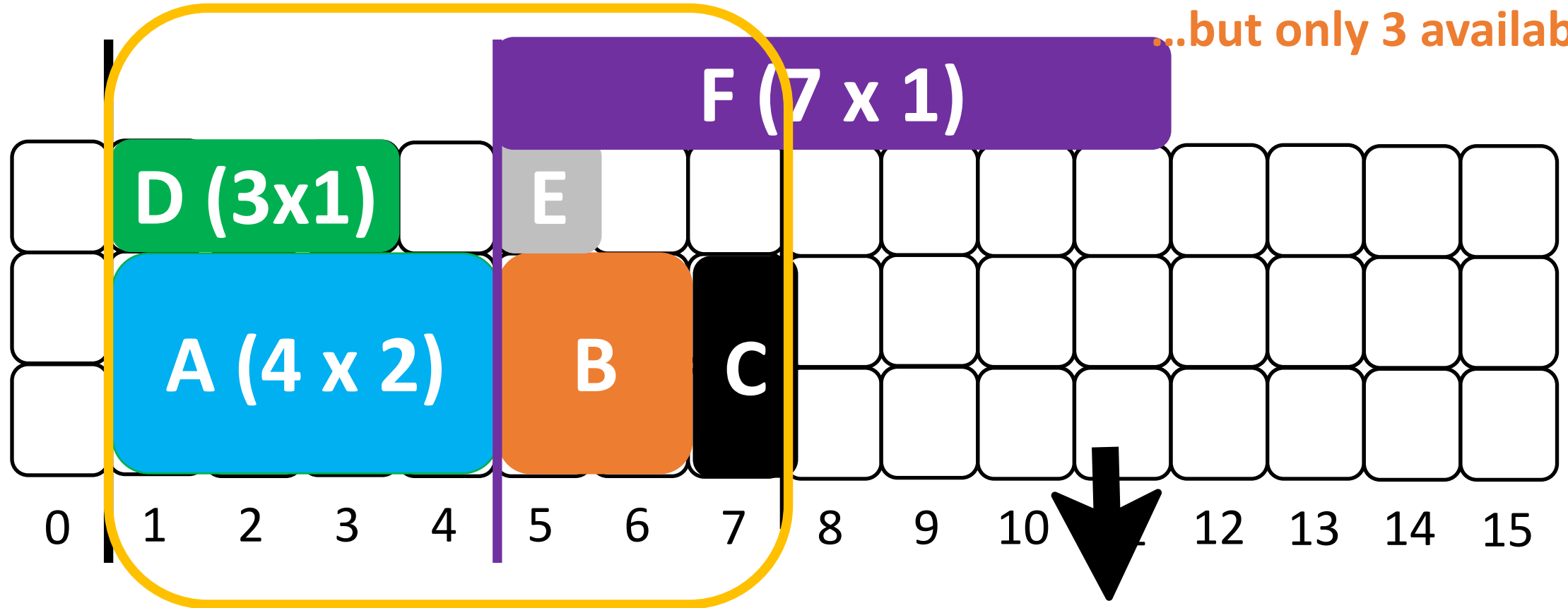


Remove time 0 from Task F!

Place task F at its earliest starting time...
How much energy would task F need from [1, 7]?

6

...but only 3 available



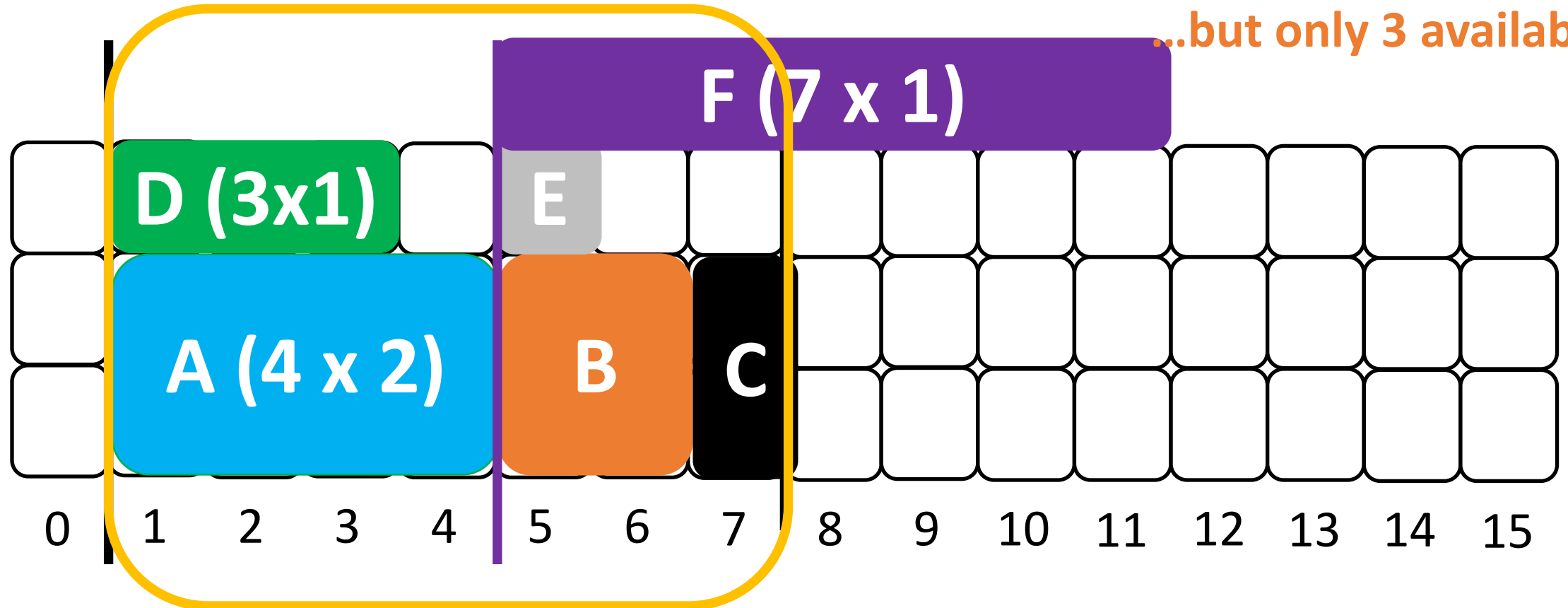
(only three energy left for task F, so it can start earliest at time 5)

Even better: $LB(\text{Task F}) \geq 5$

Place task F at its earliest starting time...
How much energy would task F need from [1, 7]?

6

...but only 3 available

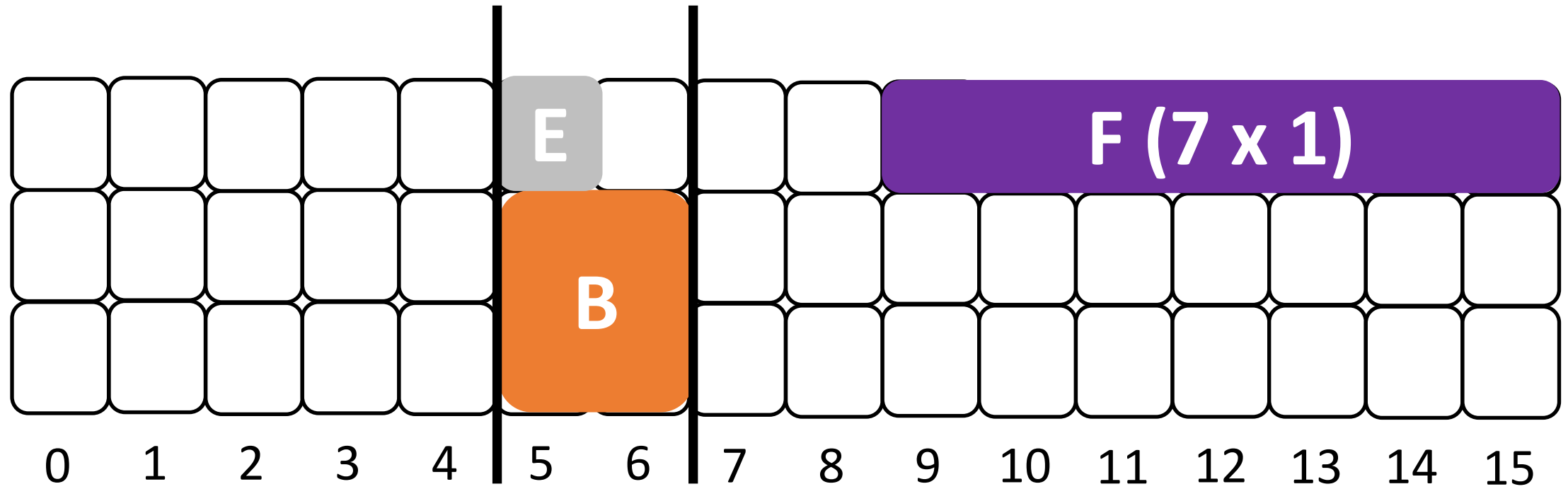


**Better than timetabling
(in this case!)**

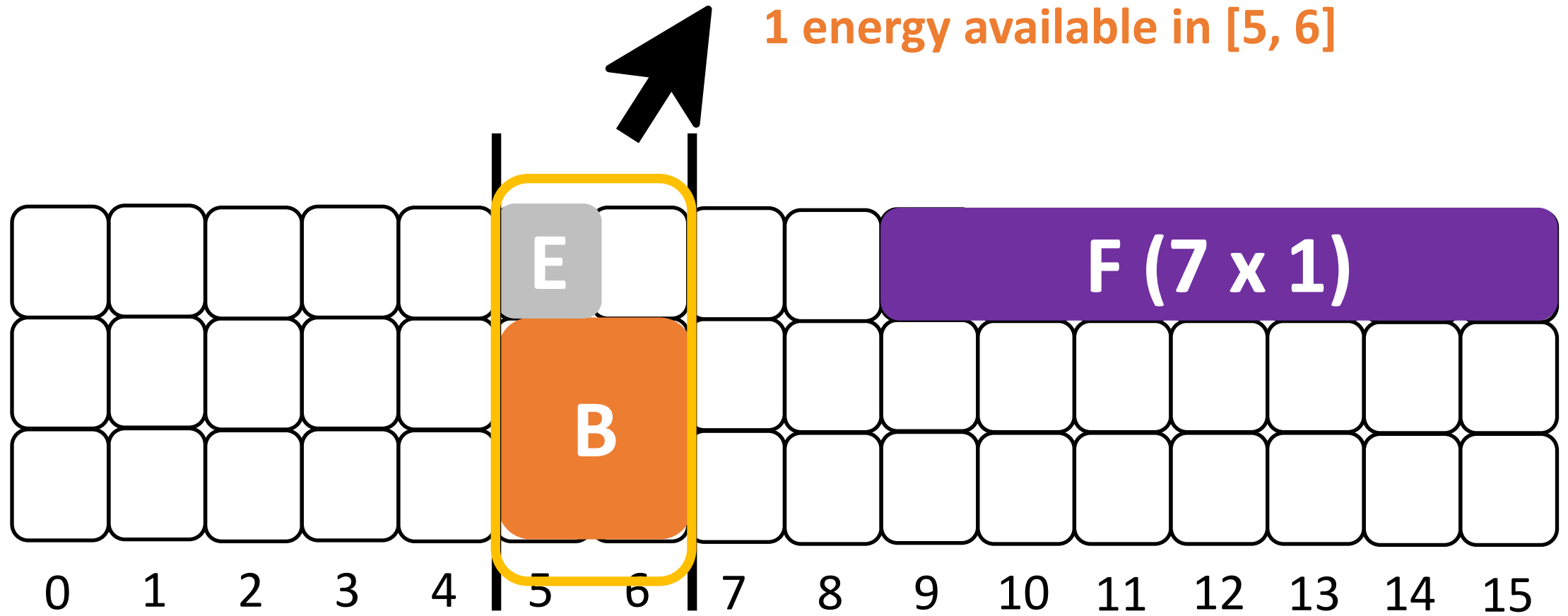
$$\text{LB}(\text{Task F}) \geq 5$$

New example

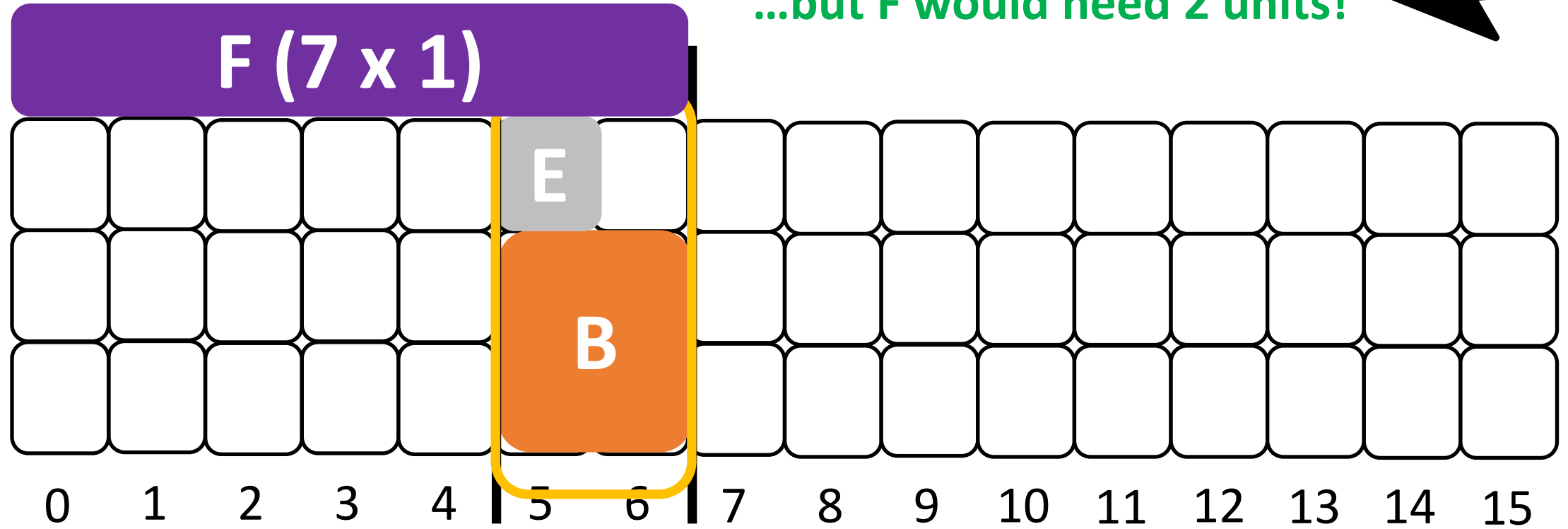
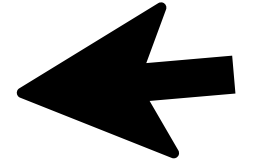
Timetabling does not lead to any propagation



5 energy required by task
1 energy available in [5, 6]



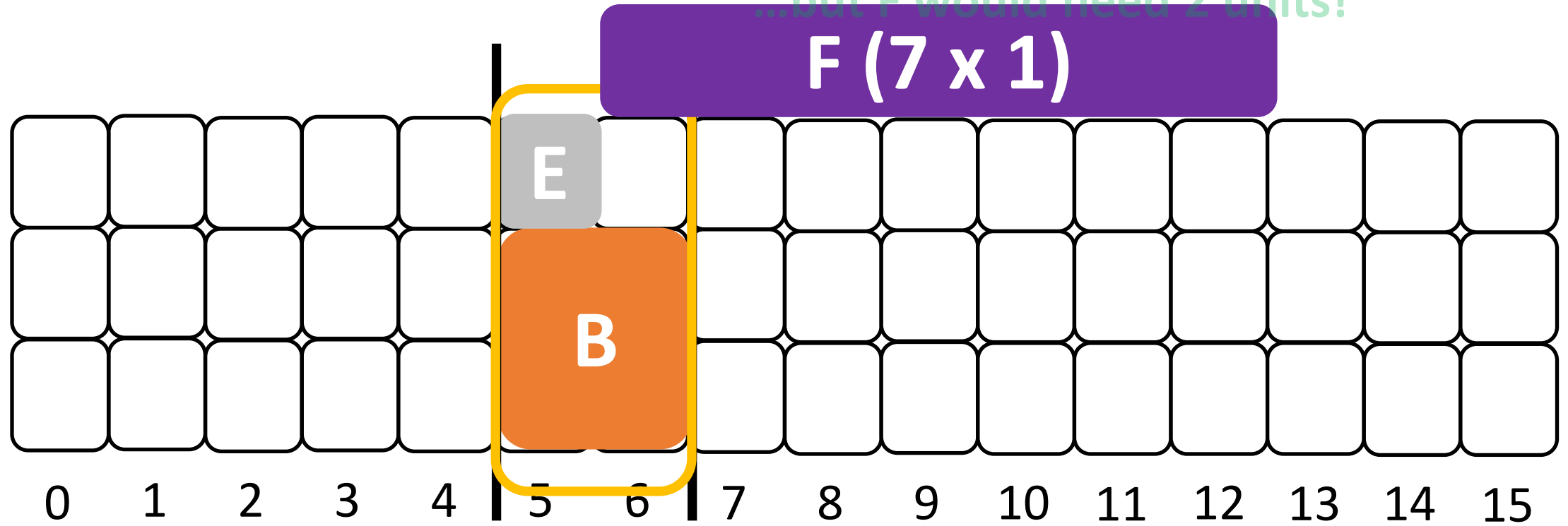
5 energy required by task
1 energy available in [5, 6]
...but F would need 2 units!



5 energy required by task

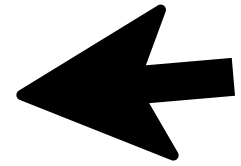
1 energy available in [5, 6]

...but F would need 2 units!



(only one energy left for task F, so it can start earliest at time 6)

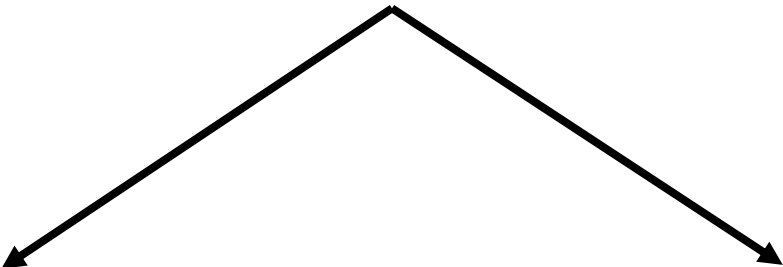
$$\text{LB}(\text{Task F}) \geq 6$$



Energetic reasoning subsumes Timetabling!
But more expensive!

In this mini-lecture...

Cumulative Constraint



```
graph TD; A[Cumulative Constraint] --> B[Timetable Reasoning]; A --> C[Energetic Reasoning];
```

**Timetable
Reasoning**

**Energetic
Reasoning**